

AQUIB ALI KHAN

Intern – Cybersecurity

Internship Focus

Practical cybersecurity projects covering password security, vulnerability assessment, phishing detection, and secure authentication.

Four practical projects • Cybersecurity • Project-based learning

Intern ID: THX-AUG1626-174

Candidate Profile

A

**AQUIB
ALI KHAN**

B.Tech – Computer Science
&
Engineering

1

Institute

IILM University, Greater Noida, U.P.

2

Programme

B.Tech CSE

3

Internship Role

Intern – Cybersecurity

4

Roll Number

25SCS1003000491

5

Work Mode

Remote / Project-Based • Practical cybersecurity projects and task-oriented learning.

Thiranex

The internship was offered by Thiranex as a remote, project-based cybersecurity internship. The work was organized around practical tasks designed to build understanding through implementation, testing, and review.

1

Role

Intern – Cybersecurity

Practical security-oriented project work.

2

Mode

Remote / Project-Based

Work organized around practical application and security tasks.

3

Mentorship

Industry-oriented guidance

Progress and project work reviewed during the internship.

From Security Concepts to Practical Solutions

Cybersecurity concepts become more useful when they are applied to practical problems. The internship addressed common security concerns through four different projects.



The internship addressed this practical gap through password security, vulnerability scanning, phishing detection, and secure login development.

What the Internship Set Out to Achieve

1

Password Security

Analyze password length, complexity, and uniqueness.

2

Vulnerability Assessment

Identify open ports, weak configurations, and outdated versions.

3

Phishing Detection

Classify emails as “Phishing” or “Safe” using machine learning.

4

Secure Authentication

Implement registration, login, password hashing, and session handling.

5

Security Practices

Understand validation and protection against common attacks.

6

Testing & Debugging

Improve testing, debugging, analysis, and problem-solving skills.

Four Practical Cybersecurity Projects

1

Password Strength Analyzer

Checks password length, complexity, uniqueness, and suggests stronger alternatives.

2

Vulnerability Scanner

Checks common security issues such as open ports, weak configurations, and outdated software versions.

3

Phishing Email Detection Model

Uses email features and Scikit-learn to classify messages as “Phishing” or “Safe”.

4

Secure Login System

Provides registration and login with password hashing, validation, session management, and security controls.

Project work was completed according to the assigned requirements and focused on practical security implementation.

The Toolkit Behind the Projects

1

Programming

Python • JavaScript

2

Web

HTML • CSS • JavaScript

3

Machine Learning

Scikit-learn

4

Security

Password hashing • Input validation • Vulnerability checks

5

Database

Database integration for user-related records

6

Development

Visual Studio Code • npm / project dependencies

7

Testing

Postman • Browser testing • Functional checks

8

Version Control

Git • GitHub

From User Input to Security Output



User → Interface → Security Processing → Data / Model → Result

The Project Development Workflow

1 Requirement Analysis

Understand project purpose, security requirements, inputs, and expected outputs.

2 Planning & Design

Plan application flow, components, data handling, and security checks.

3 Implementation

Develop the required tool, model, or authentication functionality.

4 Integration

Connect application components, data storage, and supporting libraries where required.

5 Testing & Debugging

Test different cases, identify errors, and make corrections.

6 Final Review

Check functionality, security behaviour, and whether requirements are satisfied.

Skills & Capabilities Gained

- Practical understanding of common cybersecurity problems and basic security solutions.
- Ability to evaluate password strength using defined security criteria.
- Experience with basic vulnerability scanning and security assessment.
- Hands-on practice with machine learning-based phishing email classification.
- Understanding of secure authentication, password hashing, validation, and session management.
- Improved testing, debugging, analytical thinking, and problem-solving ability.
- Experience in organizing project work and maintaining source code with Git/GitHub.
- Ability to translate security requirements into practical working solutions.

Four assigned cybersecurity projects completed as part of the internship work.

Successfully Completed



Completed Project Work

AQUIB ALI KHAN

Intern – Cybersecurity

1

01 Offer Letter

Official internship offer / onboarding document.

2

02 Project Records

Assigned tasks, project requirements, and submissions.

3

03 Communication Proof

Relevant internship communication and instructions.

4

04 Completion Evidence

Internship certificate and other supporting documents can be placed here.

THANK YOU